



GROWTERN
Academy

45-DAYS SUMMER INTERNSHIP ON AI-POWERED **PYTHON DEVELOPER**

MODULE : 01 PYTHON FUNDAMENTALS

- Introduction to Python & setup
 - Variables, data types, operators
 - Input/output operations
 - Conditional statements (if-else) & Loops (for, while)
 - Mini Task: Basic programs
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MODULE : 02 DATA STRUCTURES & FUNCTIONS

- Lists, tuples, sets, dictionaries
 - String handling
 - Functions & arguments
 - Recursion basics
 - Problem-solving exercises
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MODULE : 03 OBJECT-ORIENTED PROGRAMMING

- Classes & objects
- Constructors & methods
- Inheritance & polymorphism
- File handling (read/write files)
- Mini Project: Student Management System





MODULE : 04 **ADVANCED PYTHON & LIBRARIES**

- Exception handling
- Modules & packages
- Introduction to libraries: NumPy, Pandas
- Basics of data analysis
- Working with APIs

MODULE : 05 **PROJECTS & DEPLOYMENT**

- **Real-world projects:**
 - **Banking System**
 - **ATM Simulation**
 - **Quiz Application**
- Introduction to Git & GitHub
- Project presentation & evaluation

AI TRAINING MODULE : **GENERATIVE AI & AI TOOLS FOR DEVELOPERS**

- Introduction to Generative AI and its role in modern development
- Introduction to AI tools: ChatGPT, Google Gemini, Claude, GitHub Copilot, Perplexity, ElevenLabs, SuperBot
- Prompt engineering and effective AI usage
- Code generation using AI tools for frontend and backend development
- Building a mini project using AI-assisted workflow
- Understanding limitations, ethics, and best practices of AI

OUTCOME

Students will gain strong Python fundamentals and develop the ability to build real-world applications. They will understand object-oriented programming (OOP) concepts and data handling techniques, along with gaining hands-on project experience.

